THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

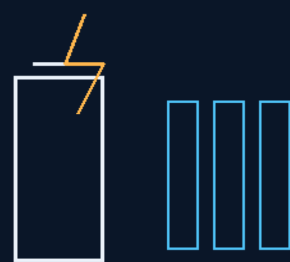
THE GEMINI PROTOCOLS

PHASE 173

FLEET BATTERY CHEMISTRY MANDATE
NiMH STANDARD, CAPACITOR BURST

Elimination beats mitigation

batteries hold energy, capacitors deliver power



backwards is the key

NiMH MOBILE · SODIUM STATIC · NO DISPOSABLES

THE CHEMISTRY LEAVES THE SHIP

Power system — v1.0 in force

GP-IM-173

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — PASS

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 173

PHASE 173: FLEET BATTERY CHEMISTRY MANDATE — NiMH MOBILE STANDARD & CAPACITOR BURST BUFFERING — STATUS: CURRENT — CLEAN PASS (VET BATCH 20). Written from seventeen years of off-grid battery life and an electrical engineering firm that built low-voltage systems — this one is the inventor's trade, not his hobby. Phase 173 mandates the fleet's portable power chemistry: every battery larger than watch-battery class is NiMH, no exceptions for convenience, weight, or fashion. The reasoning is seated whole: 100 amp-hours is 100 amp-hours — lithium's magic trick is access speed, not energy, and access speed is what starts fires; thermal runaway is a chain, not an event, and in a sealed hull elimination beats the ISS's mitigation (steel sleeves and cell screening cannot catch every cell); the industry skipped a step — lithium arrived just as NiMH hit its stride up from NiCad, and storage chemistry was abandoned mid-climb. The burst problem lithium pretended to solve is answered by physics instead of chemistry: capacitor chains deliver the surge — dual oscillating banks, one dumping while the other charges — while the battery holds the energy. Static storage stays Phase 71 sodium-ion; emergency stores run low-self-discharge NiMH; and every device aboard is rechargeable — disposable cells are prohibited cargo. The vet passed it clean.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-173, v1.0 — 23 August 2026. The one hundred and seventy-fourth manual of the program — 174 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the chemistry law as seated (contents line, core objective, the bodies — capacity illusion, fire law, the mandate, capacitors, the storage split, the prohibition — the audit and provenance, verbatim) — §2 why elimination beats mitigation (graded) — §3 the system in the cells — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 173: **Fleet Battery Chemistry Mandate — NiMH Mobile Standard & Capacitor Burst Buffering.** Prohibits thermal-runaway lithium chemistries in all cells larger than watch-battery class; standardizes crew-portable power on abuse-tolerant NiMH with chain/oscillating capacitor banks delivering lithium-class surge; extends Phase 71's zero-runaway sodium-ion doctrine to every mobile system; bans disposable cells fleet-wide and orders the open redesign of the replacement packs.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Thermal-Runaway Battery Chemistries & Disposable Cells from Crew-Portable Power *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']*'

Crew Systems Safety Baseline: Portable Power Chemistry, Burst Buffering & Disposable-Cell Prohibition

Live instruction, 30 July 2026. Written in real time with Archer, from 17 years of off-grid battery life, an electrical engineering firm that built low-voltage battery devices, and a box of dead Milwaukee packs. Physics audit appended per file doctrine.

THE CAPACITY ILLUSION — DISCHARGE RATE IS NOT ENERGY (VERBATIM)

- **100 amp-hours is 100 amp-hours.** Lithium provides no more power and no more energy. Its magic trick is access speed — the ability to dump energy fast — which creates the illusion of more overall power. On a rattle gun that trick means excess brute force: the battery dies quicker and the extra hammering is waste that destroys fasteners. A normal battery does not do this because it cannot.
- **The soda syphon rule:** put twice the pressure in a soda syphon and the glass fills faster — with blow-back and waste. Delivery pressure is not volume. Discharge rate is not capacity.
- **The inverter scandal:** lithium's genuine advantage — running to near-zero without the lead-sulfate damage that kills lead-acid at low voltage — is thrown away in houses because inverters still ship with lead-acid cut-offs (10.5V on a 12V bank, 21V on 24V). Cycle life is what destroys batteries — charge and discharge, not depth: 2000 cycles at a 10.5V cut-off or 2000 cycles at 5V.
- **Shelf truth:** lead-acid dies sitting on a service-station shelf waiting for sale. Lithium tool packs in commercial use last 1-2 years at six weeks' heavy use a year. NiMH packs in regular tool use run 4 years.

THE SEALED-HULL FIRE LAW — WHY THE CHEMISTRY LEAVES THE SHIP (VERBATIM)

- **Thermal runaway is a chain, not an event:** internal fault, separator failure, flammable electrolyte venting, then the cathode itself releases oxygen — the fire brings its own oxidizer. A CO2 extinguisher cools a surface; it cannot interrupt an electrochemical chain reaction. In a sealed hull, one cell is a catastrophe that feeds itself.

- **Elimination beats mitigation.** The ISS approach is containment — steel sleeves, chimney vents, cell screening — and even its owners admit screening cannot catch every latent defect. The fleet's answer is older and harder: remove the chemistry. NiMH's aqueous electrolyte carries no flammable organic solvent and no oxygen-releasing cathode; the fire triangle is broken at the chemistry level.

THE NiMH MOBILE STANDARD (THE MANDATE) (VERBATIM)

- **Mandate:** every battery larger than watch-battery class aboard the fleet is NiMH. No exceptions for convenience, weight, or fashion. Watch-class coin cells (Phase 166 vitals watches) are exempt — grams of mass where density is mission-critical.
- **Emergency stores use the low-self-discharge variant.** Standard NiMH sheds charge on the shelf; LSD NiMH holds it for years. Bulkhead kits, gauntlet spares, and pod equipment are LSD-specified so the kit that waits five years still works on the day.
- **Backwards is the key.** The industry skipped a step — lithium arrived just as NiMH hit its stride up from NiCad, and storage chemistry was abandoned mid-climb. The fleet goes back and takes the step properly.

CAPACITOR-CHAIN BURST BUFFERING (VERBATIM)

- **Chain or train capacitors cover fast high-power draw.** The battery holds the energy; the capacitor chain delivers the surge. Rattle-gun hammer loads, winch bursts, and tool start-up currents come off the capacitors, not the cells — lithium-class peak performance with no lithium.
- **Dual oscillating banks:** one capacitor bank dumps while the other charges, alternating — continuous high-power availability to lithium level for sustained heavy tools.
- **Gauntlet application:** the Phase 165 rewind winch runs on an NiMH pack with a capacitor-burst stage for the recovery surge. Archer's correction of record: "NiMH for the gauntlets (my fault is say lithium because its generic for small high power)."

THE STORAGE SPLIT — SODIUM STATIC, NiMH MOBILE (VERBATIM)

- **Phase 71 stands and extends:** sodium-ion remains the fleet's static storage chemistry — synthesized from salt, zero-runaway by nature. This phase extends the same zero-runaway doctrine to every mobile system: static walls run sodium, hands and tools run NiMH.
- **The Phase 51 correction is honored fleet-wide.** The emergency sled network's energetics line specified lithium blocks in direct conflict with Phase 71's *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: 'absolute']* prohibition within that same network. Phase

71 wins; the sled runs sodium-ion. The conflict and its correction are kept visible in Phase 51's markup — flags correct, never silently delete.

DISPOSABLE-CELL PROHIBITION & THE REDESIGN ORDER (VERBATIM)

- **No primary cells aboard.** Every battery-powered device in the fleet is rechargeable. Disposable batteries are prohibited cargo — dead weight, fire cargo, and a supply-chain dependency in one package.
- **The redesign order:** a suitable open-source replacement pack is to be designed and published in this file — NiMH cell geometry, capacitor-burst stage, and mechanical interface — so any colony workshop can build crew power from first principles. Likely easier made on a new planet colony than lithium mining ever was: nickel is among the most abundant metals in iron-nickel asteroids; sodium comes from salt; lithium is cosmically scarce.

PHYSICS NOTE — THE AUDIT, BOTH DIRECTIONS (VERBATIM)

The safety case is settled physics. Thermal runaway's chain — fault, separator failure, electrolyte vent, cathode oxygen release — is documented to the point that space agencies package lithium cells in steel sleeves with chimney vents and still concede that screening misses latent defects. A fire that manufactures its own oxidizer cannot be smothered; CO_2 and halon-class agents cool surfaces while the electrochemistry continues underneath. NiMH's aqueous potassium-hydroxide electrolyte removes both the flammable solvent and the oxygen source. The capacity illusion is dimensionally exact: amp-hours are charge, and C-rate changes the peak, not the total — the soda syphon analogy is the correct model.

The honest trades stay in the file, because the no-invention doctrine cuts both ways. Published laboratory figures favor lithium on cycle count (roughly 1,000-6,000 versus 500-1,000), energy density (150-250 versus 60-120 Wh/kg), and self-discharge (1-3%/month versus up to 30%). The mandate absorbs these costs deliberately: the mass penalty is trivial at handheld scale, LSD NiMH (about 10% loss per year) covers the emergency stores, and Archer's field numbers — four-year NiMH tool packs against one-to-two-year lithium packs under commercial abuse — measure what laboratory cycling does not: heat, overcharge, hammer loads, and neglect. Both sets of numbers are true in their own conditions; the fleet optimizes for the conditions it will actually have. The capacitor architecture is standard hybrid power engineering — supercapacitors for the kilowatt surge, cells for the watt-hours — no new physics required. One manufacturing caveat is flagged for the colony workshops: classic AB5 hydride alloys need lanthanide mischmetal, which is trace in asteroids, so colony cells should plan on AB2 titanium-zirconium hydride alloys from more common refractory metals. The mandate stands; the alloy recipe is the colony's homework.

ORIGIN OF THOUGHT — PHASE 173 (VERBATIM)

Archer, 30 July 2026: "batteries are ingrained in my lifestyle and history, lived off grid 17 years, own everything rechargeable worked with every hand held tool, ran an electrical engineering firm that built low voltage devices and utilized batteries. the lithium story is 50 percent myth, firstly a 100 amp hour battery is just that, lithium does not provide more power at all still 100amp hours. its ability to access the energy at speed to dump energy is its magic trick that gives the illusion of more overall power, but it simply mean the battery in a power tool dies quicker, notably on things like a rattle gun, because it uses excess brute force, a normal battery does not because it cannot. put twice the pressure in a soda syphon, and i will fill the glass faster but there will be blow back and waste, the extra psi hammering is waste and destroys more fasteners."

Archer, 30 July 2026: "lithium batteries can go down to near zero without damage to the battery, the lead sulfates buildup that occurs in lead acid at low volage stress does not occur. stupidly this wonderful get more energy story which is true isn't used in houses, because the inverters all still run 10.5 low voltage cut off at 12 volts 21v at 24 volts etc. you will kill the battery quicker they yell, eerr no, you will, batteries have cycle life, that's what destroys the, charge and discharge, 2000 cycles at 10.5v or 2000 cycles at 5 volts cuttoff. lead acid will die sitting on a shell in a service station for years waiting for sale, lithium batteries dont last 20 years in power tools for commercial use, 1-2years max and only 6 weeks heavy use each year. have killed plenty of Milwaukie batteries on rattle guns."

Archer, 30 July 2026: "So reality and safety, backwards is the key, NiMH , is my preferred battery very long life in use 4years for power tools used regularly. we made the mistake of not going to storage because lithium came along just as it hit stride up from NiCad. their capacity for low discharge is also great, chain or train capacitors would cover fast high power draw usage, even dual oscillating capacitors would cover most things to lithium level. So whilst we are going Sodium Ion for storage, we are switching out as a mandate that all lithium batteries larger than a watch battery, be switched out for NiMH. likely more easily made on a new plant colony the lithium mining. in any event we will redesign a suitable battery replacement on here to remove disposable batteries."

Why it matters, in plain English: a fire that brings its own oxygen cannot be put out — it can only be prevented. Every other safety device in this file assumes the crew can fight the emergency; a lithium fire in a sealed hull is the one emergency that fights back and wins. So the fleet reaches backwards for the chemistry that forgives, and lets capacitors do the showing off.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CAPACITY ILLUSION, AFFIRMED: amp-hours are energy; discharge rate is power. Lithium provides no more of the first and its advantage in the second is precisely the hazard — the soda-syphon rule: twice the pressure fills the glass faster, with blow-back and waste. Delivery pressure is not volume.

THE FIRE LAW, AFFIRMED BOTH DIRECTIONS: thermal runaway is a chain — internal fault, separator failure, flammable electrolyte venting, then the cathode releases its own oxygen and the fire brings its own oxidizer. In a sealed hull you cannot vent the chain, so the chemistry leaves the ship. Elimination beats mitigation: even the ISS's own documentation admits screening cannot catch every cell.

THE CAPACITOR ANSWER, AFFIRMED: batteries hold energy, capacitors deliver power — dual oscillating banks give continuous high-power availability to lithium level for sustained heavy tools, with no flammable electrolyte anywhere in the surge path. The Phase 165 gauntlet rewind winch is the seated application.

THE SKIPPED STEP, AFFIRMED AS INDUSTRIAL HISTORY: NiMH was still climbing when lithium arrived — the mandate is not nostalgia, it is resuming an abandoned development line with decades of field data behind it, from a man who lived off-grid on batteries for seventeen years.

§3 — THE SYSTEM IN THE CELLS

- THE MANDATE: every cell larger than watch-battery class is NiMH — no exceptions for convenience, weight, or fashion.
- THE EMERGENCY STORES: low-self-discharge NiMH — bulkhead kits, gauntlet spares, pod kits hold charge for years.
- THE BURST STAGE: capacitor chains, dual oscillating banks — one dumps while the other charges.
- THE SPLIT: sodium-ion static (Phase 71 stands), NiMH mobile — each chemistry where its mass belongs.
- THE PROHIBITION: no primary cells aboard — every device rechargeable; disposables are dead weight and fire cargo.
- THE REDESIGN ORDER: a suitable open-source replacement pack designed and published in this file — NiMH geometry, capacitor-burst stage, mechanical shell.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE ELIMINATION DOCTRINE: the hazard that cannot be mitigated is removed — the chemistry leaves the ship.
- THE BACKWARDS-IS-THE-KEY DOCTRINE: when the industry skips a step, the fleet goes back and takes it.
- THE ENERGY-POWER SPLIT DOCTRINE: batteries hold energy, capacitors deliver power — never ask one device to be both.

- THE NO-DISPOSABLES DOCTRINE: everything aboard recharges — the closed loop extends to the cells.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 20 (15 August 2026 — phases 161-175, cross-checked in sitting 307), SEATED. The grade, verbatim from the batch: '173 → PASS.' The ledger line: '173 PASS.' No red, no yellow — a clean green on the fleet's fire law. The 15 August blanket wording kill ('100%' removed from the core objective) is carried inline in §1.

§6 — BUILD SEQUENCE

- STEP 1 — Standardize the pack geometry: NiMH cell formats and shells for every device class aboard.
- STEP 2 — Bench the capacitor chain: dump/charge alternation, rattle-gun hammer loads, winch surges.
- STEP 3 — Retrofit the manifest: every lithium device redesigned to the NiMH pack — the redesign order executed.
- STEP 4 — Stock the LSD stores: emergency kits sealed and dated.
- STEP 5 — Publish the open-source replacement pack — cell geometry, burst stage, shell — under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 71 — sodium-ion static storage: the split's other half; the zero-runaway wall chemistry.
- PHASE 165 — the tether gauntlets (GP-IM-165): the rewind winch runs NiMH with the capacitor-burst stage — his correction of record seated.
- PHASE 174 — flow-gel inline recharge (GP-IM-174): the NiMH standard's charging infrastructure.
- PHASE 51 — the emergency sled network: the lithium conflict corrected fleet-wide per this mandate.

§8 — DO-NOT-BUILD

- DO NOT ship lithium — no thermal-runaway chemistry in any cell larger than watch class.
- DO NOT buy the capacity illusion — discharge rate is not energy; spec the surge to capacitors.
- DO NOT carry primary cells — rechargeable everything; disposables are prohibited cargo.
- DO NOT mitigate what can be eliminated — sleeves and screening are the ISS's compromise, not ours.

§9 — OWED

OWED: nothing flagged — a clean green. Standing duty: the open-source replacement pack publishes from the bench (§6). The 15 August blanket wording kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-173, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The one hundred and seventy-fourth manual of the program — 174 of 290. Built 23 August 2026, sixth of the 20-manual 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'. The phase's own chain, all seated in the master: the contents line, the masthead trio and the four bodies (as seated); the live instruction of 30 July 2026 — seventeen years off-grid, the electrical firm, the trade not the hobby; the 15 August blanket wording kill ('100%' struck from the core objective) carried inline; the vet batch 20 clean pass in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the replacement-pack drawings, or his word, or his word.

END OF MANUAL — PHASE 173